Paleomagnetism and Tectonics of a Deccan Trap Lava Sequence at Amarkantak, India[1]

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Detailed paleomagnetic measurements have been made on a lava sequence exposed between altitudes of 700 to 1000 meters in the Amarkantak-Dindori tract on the eastern margin of the Deccan traps. The variation in the direction of remanent magnetism with altitude has been studied. The paleomagnetic data reveal the occurrence of a reversal of the earth's magnetic field at an altitude of about 975 meters at Amarkantak as well as the unstable behavior of the geomagnetic field near the polarity transition. These data are at variance with the observation of the same reversal of magnetic field at an altitude of about 625 meters in the paleomagnetic study of several other vertical sequences in the Deccan traps. The variance is interpreted in terms of an uplift of the Amarkantak region by about 350 meters.

INTRODUCTION

The Deccan traps cover a large areal extent of the Indian peninsula, as can be seen from Figure 1. This volcanic activity is assigned an age span of Upper Cretaceous to Eocene. It is supposed to have begun in the east and extended westward up to the coast, where the traps are believed to have attained maximum thickness.

The early paleomagnetic work on this formation [*Clegg et al.*, 1956; *Deutsch et al.*, 1959] was more or less restricted to the ridges running parallel and close to the western coast. Since these studies were made, additional paleomagnetic data have been obtained, and the Deccan traps are now regarded as one of the most extensively studied of such formations [*McElhinny*, 1968]. Paleomagnetic data have been obtained from a large number of samples collected from numerous sites, some of which are hundreds of miles apart.

In these studies of vertical sequences in the Deccan traps, a rather unique feature is revealed at several points: lava flows that are below an approximate altitude of 625 meters are reversely magnetized (azimuth is nearer to 180° and positive dip), whereas flows above this altitude are normally magnetized (azimuth is nearer to 360° and negative dip). *Deutsch et al.* [1959] were the first to make this observation for the western ridges, and it has been subsequently confirmed by work in other regions of the Deccan traps [*Sahasrabudhe*, 1963; *Verma & Pullaiah*, 1970].

This polarity transition in the Deccan traps can be used as a marker for stratigraphic correlation amongst the flows and also for studying any post-Deccan trap tectonic movements such as block uplift and subsidence.

GEOLOGY OF THE AREA

The Amarkantak hill is a prominent feature located on the northeastern margin of the Deccan traps and is the source point for the river Narmada. The traps in the Jabalpur-Dindori-Amarkantak area are believed to belong to the early phases of eruption [*Pascoe*, 1964] of the Deccan lavas. In Figure 1 this area of the Deccan traps has been distinguished by dotting. A detailed map of the area is given in Figure 2.

The Amarkantak plateau has a height of about 1050 meters. The average height of the region immediately to the east of the hill is about 600 meters, whereas the terrain on the west slopes gently up to about 600 meters near Dindori, about 65 km from Amarkantak.

A geological section of the Amarkantak hill, based on the description by *Fermor* [1916], is

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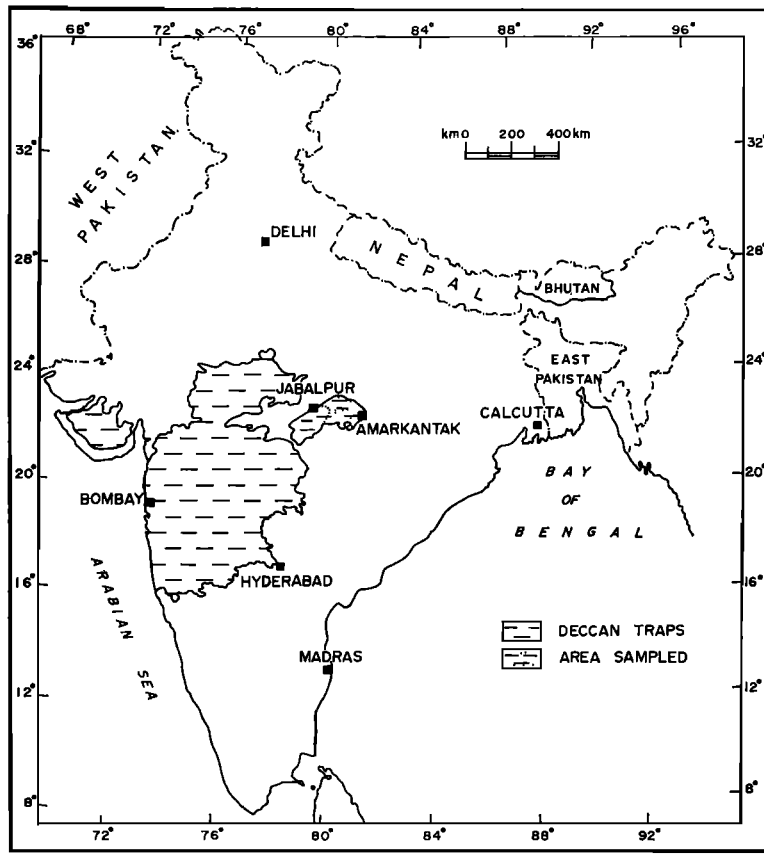


Fig. 1. Outline map of India showing the areal extent of the Deccan traps.

shown in Figure 3. A rather peculiar feature of the hill is that it is completely covered on the western side by a thick succession of basaltic flows that also form the top 250 meters of the hill. Ascending the hill from the eastern side, however, one comes across basement gneisses up to an altitude of 750 meters. These are overlain by about 30 meters of sediments belonging to the Talchir series of Upper Carboniferous age and another 30 meters of Lameta sediments of Cretaceous age that are covered by the Deccan traps.

A vertical section of about 300 meters of lavas was sampled between altitudes 700 to 1000 meters in the Amarkantak-Dindori tract shown in Figure 2. In total, 19 sites were sampled and about 120 samples were collected; of these, the samples from one site (site 8) were lost in transit.

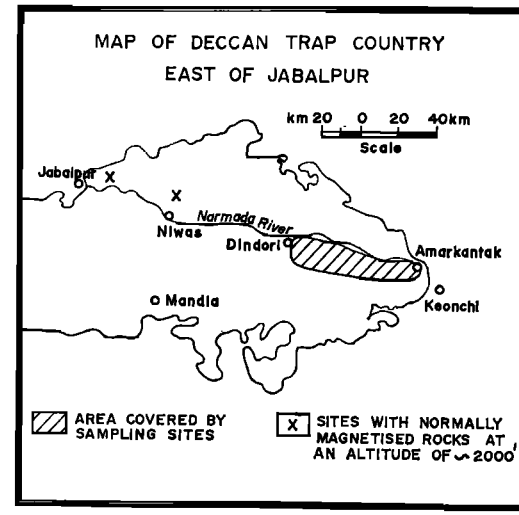


Fig. 2. Detailed map of the Deccan traps east of Jabalpur.

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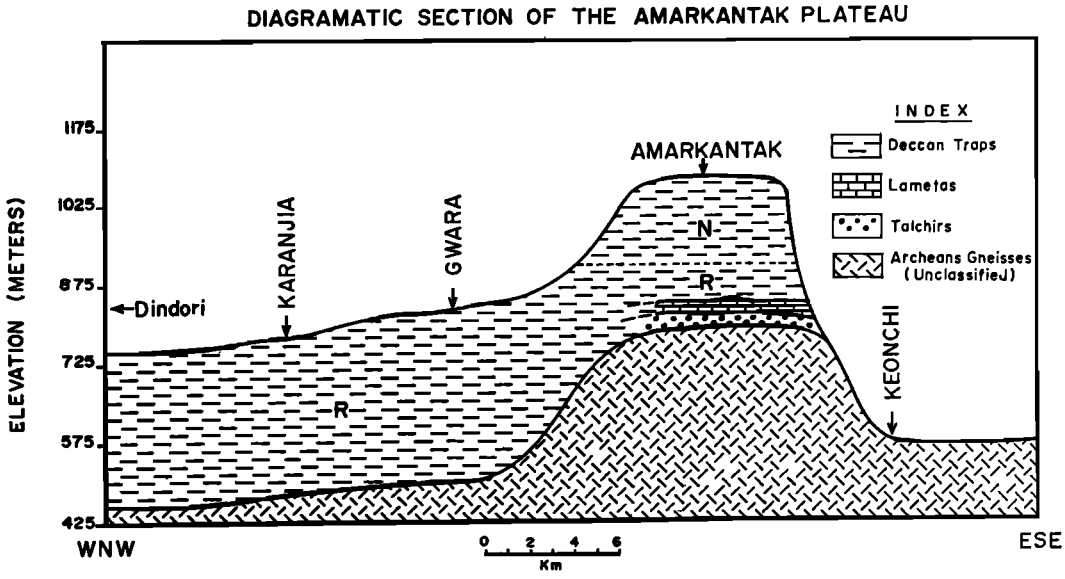


Fig. 3. Geological section at Amarkantak.

Paleomagnetic Results

Three or four specimens were cored from each sample, and their NRM directions were measured. A few pilot specimens from each site were subjected to alternating field demagnetization in successively increasing peak-field strength of 25, 50, 75, 100, 150, 200, and 300 oe. On the basis of the change in directions and the drop in intensity of the pilot specimens, a peak-field value was fixed for batch cleaning of the remaining specimens from each site.

The specimens from all the sites except site 1 were sufficiently cleaned by alternating field (AC) demagnetization, and the scatter in the NRM directions for each site was considerably reduced. In the case of site 1, the scatter in the directions of specimen from each sample was considerably reduced by AC cleaning, but the scatter among the mean sample directions from this site remained rather too large. This was attributed to the rather doubtful in situ condition of the spheroidal boulder samples coming from this site, and the results for the site were therefore rejected.

On AC cleaning, most of the sites were found to be reversely magnetized, whereas the two sites at the top were found to be normally magnetized. Sites 5 and 14–17 gave results that could best be classified as intermediate. To ascertain that the apparently anomalous directions from these sites were primary TRM directions, the AC-cleaned specimens from these five sites were subjected to thermal demagnetization up to different temperature levels and to cooling in the zero field [*Athavale*, 1969]. The stable paleomagnetic directions for all the sites are given in Table 1, along with other details such as altitude, number of samples, *Fisher's* [1953] coefficient of scatter, virtual geomagnetic pole position, and stability tests employed in each case.

Morphology of Reversal of Earth's Magnetic Field

Figure 4 shows the variation in declination and inclination of the remanent magnetic vector with the altitude of each site. The directions in Table 1 can be classified in three groups: (1) sites 2–4 and 6–13 (701–953 meters), reverse; (2) sites 14–17 (959–981 meters), intermediate; (3) sites 18–19 (990–999 meters), normal.

Figure 5 shows the migration of the latitude of the virtual geomagnetic pole from southern to northern hemisphere as a result of the reversal of the earth's magnetic field.

Site 5 gives an anomalous direction, i.e., it gives normal azimuth but positive dip. The primary character of this magnetization has

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been ascertained by AC cleaning up to 300 oe and by thermal demagnetization up to 350°C. The directions from individual specimens form a close cluster, which is maintained after AC and thermal cleaning at higher peak fields and temperatures, respectively. The following interpretations are possible for the directions of this site.

1. The samples are reversely magnetized, and the site has rotated through about 140° subsequent to acquisition of the TRM. However, since this site is sandwiched within a succession of flows giving reversed magnetization without any indication of local structural rotation, such an explanation does not seem to be tenable.

2. The samples could have acquired secondary magnetization because of lightning strike. The NRM intensity of these samples is comparable to that of other nearby sites. A lightning-induced remanent magnetization would be distinguished by very high intensity. In the absence of such indication from the intensity of the samples of site 5, this possibility is ruled out.

3. It is theoretically presumed [*Elsasser*, 1956] that during a reversal the strength of the dipole wanes to a minimum and waxes again in the opposite direction after the flip-over. Under such circumstances, the nondipole components become predominant, and a resultant field direction far removed from that for the dipole configuration is possible. (The stable directions of sites 14–17 can be explained as being caused by such a phenomenon since they are followed by evidence of polarity transition).

However, even this explanation cannot be applied to the case of the discordant direction for site 5 because of the reason given above, i.e., the site is sandwiched in a succession of flows that do not give any anomalous or intermediate directions and that clearly belong to one polarity group.

4. The direction given by the site represents a precursor to the main event of reversal. In other words, the direction records an incomplete reversal of the field, i.e., the field was about to be reversed but returned to its original state.

Watkins [1969] has recently reported a similar observation in the case of the remanent magnetic directions of a sequence of 77 Miocene lava flows from the Steens mountain range of Oregon, U.S.A. In this case, the actual polarity transition from normal to reverse stage takes place near about the fifty-seventh flow and is preceded by intermediate directions in eight to

TABLE 1. Paleomagnetic Data for Amarkantak (22°40′N; 31°45′E)

Site No.	No. of Samples	Altitude, meters	Mean Direction		α 95, deg	Virtual Geomagnetic Pole		Stability Tests
			Decl.	Incl.		Lat.	Long.	
2	7	701	146	+46	4.3	31°S	118°E	A.C. 100 oe
3	7	731	145	+43	4.3	32°S	120°E	A.C. 100 oe
4	7	795	143	+38	10.	33°S	124°E	A.C. 100 oe
5	5	825	321	+35	3.1	53°30′N	8°W	A.C. 150 oe; T.D. 350°C
6	5	856	138	+43	9.3	27°S	125°E	A.C. 100 oe
7	6	868	151	+56	4.4	30°S	109°E	A.C. 100 oe
9	6	877	149	+50	3.6	29°S	112°E	A.C. 100 oe
10	3	898	149	+42	10.	35°S	117°E	A.C. 100 oe
11	8	937	162	+51	5.	32°S	99°E	A.C. 75 oe
12	4	946	167	+51	4.	30°S	95°E	A.C. 75 oe
13	4	953	155	+35	5.7	46°S	117°E	A.C. 75 oe
14	5	959	221	+51	3.1	23°S	46°E	A.C. 100 oe; T.D. 170°C
15	7	966	220	+53	1.	22°S	50°E	A.C. 75 oe; T.D. 150°C
16	6	975	47	+36	3.5	46°N	163°E	A.C. 100 oe; T.D. 200°C
17	6	981	45	+54	5.4	47°30′N	142°E	A.C. 100 oe; T.D. 250°C
18	7	990	328	−31	6.1	40°N	55°W	A.C. 75 oe
19	5	999	326	−38	5.4	36°N	58°W	A.C. 75 oe

Note.
AC = alternating field demagnetization; TD = thermal demagnetization.

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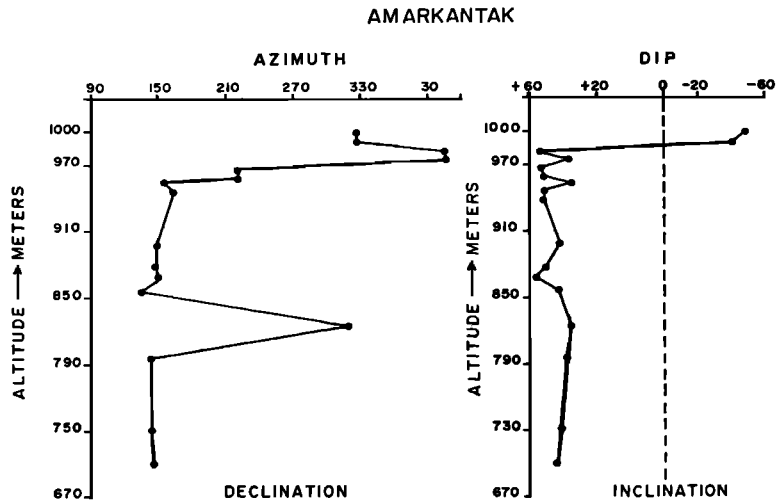


Fig. 4. Variation of declination and inclination with altitude of sampling sites.

ten flows, which is similar to our case of sites 14–17. However, well within the group of normally magnetized rocks, Watkins reports directions of flows 28 and 29 as showing reversed azimuth and normal dip, which is analogous to the case of site 5 in the Amarkantak sequence.

Watkins [1969] cites one more case where a similar observation has been made and infers that in view of their proximity in time to the true polarity transition, these anomalous directions imply a stage of dipolar instability associated with the reaching of threshold conditions for a flip-over of the field. This short excursion of the field prior to its main reversal has been named by Watkins as a 'rebound effect,' which is expected on theoretical grounds since it is consistent with the analog solutions obtained by *Mathews and Gardner* [1963] for the output of *Rikitake*'s [1958] double-disk dynamo model of the geomagnetic field.

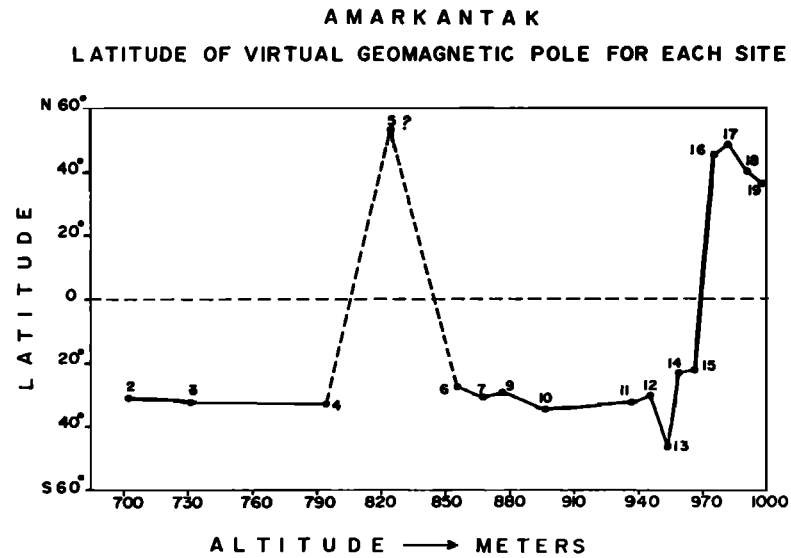


Fig. 5. Migration of the latitude of virtual geomagnetic pole with altitude of sampling sites.

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This, therefore, appears to be the most plausible explanation for the direction given by site 5.

Tectonics of Amarkantak Region as Interpreted from Paleomagnetic Data

Verma and Pullaiah [1970] have studied the paleomagnetism of trap flows between Dindori and Jabalpur and have located polarity transition at two points near an elevation of about 625 meters (Figure 2). The difference of about 350 meters in the polarity-transition levels at Amarkantak and Jabalpur could be attributed to one of the following possibilities.

1. Occurrence of more than one reversal of the earth's magnetic field during eruption of the Deccan traps.

2. A comparatively rapid piling of an additional 350 meters of lava flows at Amarkantak during the same period over which the Deccan traps elsewhere had attained a height of about 625 meters.

3. Uplift of the Amarkantak region subsequent to the cessation of Deccan trap activities.

Geological evidence does not favor either the first or the second hypothesis. The oldest Deccan trap flows are found to be overlying Cretaceous Lameta sedimentary beds in various sections of the Deccan traps, including those in the Jabalpur region [*Wadia,* 1961]. As shown in Figure 3, the traps at Amarkantak also overlie the Lameta beds and should therefore be considered as contemporaneous to those formed by the earliest outpouring of lavas in other Deccan trap regions. Further, the oldest Deccan traps overlying Lameta beds are exposed in Jabalpur region at an altitude of about 400 meters, thus giving a thickness of about 225 meters for the reversed traps below the polarity transition. This thickness compares very well with the thickness of reversed traps between the Lametas and the polarity transition at Amarkantak (see Figure 3).

It therefore appears quite certain that the polarity transition at Amarkantak at about 975 meters is the same transition that is met with in the Deccan trap country elsewhere at an altitude of approximately 625 meters.

Conclusions. Detailed paleomagnetic study of a 300-meter thick sequence of Deccan Trap flows in the Amarkantak region has led to the following conclusions.

1. Polarity transition from reversely magnetized to normally magnetized traps occurs at Amarkantak at an altitude of about 975 meters, whereas in Deccan trap sections elsewhere it occurs at an altitude of 625 meters.

2. The record of paleomagnetic directions near the polarity transition offers evidence for the theoretically expected 'rebound effect,' or short excursions of the dipole prior to the main reversal of the field.

3. The altitude of the polarity transition in the Amarkantak sequence suggests that this region has been uplifted by about 350 meters in relation to other Deccan trap covered regions exposed in the Narmada river basin.

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